Prob. 4. (Z&T, Ch. 6, 1.4)

Let $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$, Ω = reg. dom. & suppose that it is a <u>nontrivial</u> sol-n (the trivial sol-n is $u=0$) of

(1) $$\underline{\nabla}^2 u + \lambda u = 0 \quad \Omega$$

(2) $$u = 0 \quad \text{on } \partial\Omega.$$

where λ = const. Show that $\lambda \geq 0$.

(Hint: Use Green's identities with $w = u$.)

Sol-n. Recall the Green's 1st identity

(3) $$\int_\Omega u \cdot \underline{\nabla}^2 w \, d\underline{x} = \int_{\partial\Omega} u \cdot \frac{\partial w}{\partial \underline{n}} \, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, d\underline{x}$$

with $\boxed{w = u}$, (3) becomes:

$$\int_\Omega u \cdot \overbrace{\underline{\nabla} u^2}^{= -\lambda u \text{ by (1)}} \, d\underline{x} = \int_{\partial\Omega} \overset{0 \text{ by (2)}}{u} \cdot \frac{\partial u}{\partial \underline{n}} \, ds - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} u \, d\underline{x} \quad \Rightarrow$$

$$\int_\Omega u \cdot (-\lambda u) \, d\underline{x} = \int_{\partial\Omega} 0 \, ds - \int_\Omega (\underline{\nabla} u)^2 \, d\underline{x} \quad \Rightarrow$$

$$-\lambda \int_\Omega u^2 \, d\underline{x} = -\int_\Omega (\underline{\nabla} u)^2 \, d\underline{x} \quad \Big\} * (-1) \text{ of BS:}$$

(4) $$\lambda \cdot \underbrace{\int_\Omega u^2 dx}_{>0} = \underbrace{\int_\Omega (\underline{\nabla} u)^2 d\underline{x}}_{\geq 0}$$

[note $u \neq 0$ (nontrivial sol-n) $\Rightarrow$ $u^2 \neq 0$ & $\int_\Omega u^2 \, d\underline{x} > 0$ & $\int_\Omega (\underline{\nabla} u)^2 d\underline{x} \geq 0$ as $u^2 > 0$, $(\underline{\nabla} u)^2 \geq 0$]

$\Rightarrow$ Thus for equality (4) to hold, $\boxed{\lambda \geq 0}$. ∎

If $\lambda < 0$

(5) $$(-) \cdot (+) = (+) \quad ↯ \text{ (which is impossible)}$$